La Scuola d'Italia / Huson / Physics
26 January 2026

First & last name: SOLUTIONS
Grade:

2.3 PreQuiz: Motion and Kinematics

1. Round each value to three significant figures.

   (a) 0.005283 m $\approx 0,00528$ m

   (b) 47.652 kg ≈ 47.7 kg

   (c) 8.0074 s ≈ 8.01 s

   (d) 15,340 N $\approx 15,300$ N

2. Write each number in proper scientific notation ($1 \leq a < 10$).

   (a) 0.000628 $= 6.28 \times 10^{-4}$

   (b) 72,500
   $= 7.25 \times 10^{4}$

   (c) 304,000,000 $= 3.04 \times 10^{8}$

   (d) 0.00915 $= 9.15 \times 10^{-3}$

3. Perform each operation and give the answer to 3 significant figures.

   (a) $(4.8 \times 10^2) + (5.2 \times 10^3)$
   $= 24.96 \times 10^{5}$
   $\approx 2.50 \times 10^{6}$

   (b) $(7.20 \times 10^{-3}) \times (4.0 \times 10^4)$
   $= 2.88 \times 10^{2}$

   (c) $\dfrac{8.4 \times 10^6}{2.1 \times 10^3} = 4.00 \times 10^{3}$

4. Convert each value. Show one line of work using unit factors.

   (a) 4.75 m to cm
   $4.75 \text{ m} \times \dfrac{100 \text{ cm}}{1 \text{ m}} = 475 \text{ cm}$

   (b) 2.80 km to m
   $2.80 \text{ km} \times \dfrac{1000 \text{ m}}{1 \text{ km}} = 2800 \text{m}$

   (c) 0.450 kg to g
   $0.450 \text{ kg} \cdot \dfrac{1000 \text{ g}}{1 \text{ kg}} = 450 \text{ g}$

   (d) 6.30×10^4 g to kg
   $6.30 \times 10^4 \text{ g} \times \dfrac{1 \text{ kg}}{10^3 \text{ g}}$
   $= 63.0 \text{ kg}$

5. Convert each speed.

   (a) 72.0 km/h to m/s

$$72.0\ \frac{km}{hr} \times \frac{1000\ m}{1\ km} \times \frac{3600\ s}{1\ hr} = \cancel{259{,}200{,}000}$$

20.0 m/s

   (b) 8.50 m/s to km/h

$$8.50\ \frac{m}{s} \times \frac{1\ km}{1000\ m} \times \frac{3600\ s}{1\ hr} = 30.6\ km/hr$$

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6. A student walks along a straight hallway (forward is +). Start $x_0 = 3.0$ m.
   (a) She walks to $x = 11.0$ m. Find displacement Δx.

   11.0 - 3.0m = 8.0 m

   (b) From $x = 11.0$ m she walks back to $x = 5.0$ m. Find Δx.

   5.0m - 11.0 = -6.0m

   (c) Find total displacement from start to finish.

   5.0m - 3.0m = 2.0M

   (d) Find total distance traveled.

   |8.0m| + |-6.0m| = 14.0m

7. A cart moves from $x_1 = 2.0$ m at $t_1 = 0$ s to $x_2 = 8.6$ m at $t_2 = 4.0$ s.
   (a) Find displacement. 8.6m - 2.0m = 6.6m

   (b) Find average velocity v_{avg}.

   $\frac{6.6m}{4.0-0.0s} = 1.65$ m/s ≈ 1.7 m/s

8. One-dimensional vector additions: find the total displacement.
   (a) $\Delta x_1 = +18$ m, $\Delta x_2 = -7$ m
   (b) +2.30 km, −0.80 km, +1.50 km

   +11 m

   3.00 km

Kinematics Equations

Formulas: $v = v_0 + at$ $\quad d = v_0 t + \frac{1}{2}at^2$ $\quad v^2 = v_0^2 + 2ad$ $\quad d = \frac{1}{2}(v_0 + v)t$

9. How far can a cyclist travel in 2.5 h along a straight road if her average velocity is 18 km/h?

$$d = 18\ \frac{km}{h} \cdot 2.5\ h = 45\ m$$

10. A car accelerates uniformly from rest at 2.4 m/s^2.

(a) What is its velocity after 6.0 s?

$$v = 2.4\ \frac{m}{s^2} \cdot 6.0\ s = 14.4\ \frac{m}{s}$$

$$\approx 14\ \frac{m}{s}$$

(b) How far does it travel during this time?

$$d = \frac{1}{2}\left(2.4\ \frac{m}{s^2}\right)(6.0\ s)^2$$

$$= 43.2\ m$$

11. A train traveling at 25 m/s applies its brakes and decelerates at 1.5 m/s^2.

(a) How long does it take for the train to stop?

$V = V_0 + at$

$V = 25 \frac{m}{s} + (-1.5 \frac{m}{s^2}) t = 0$

$t = \frac{25 \ m/s}{1.5 \ m/s^2} = 16\frac{2}{3}$ seconds

(b) What distance does the train travel while braking?

$d = \frac{1}{2}(V_0 + V) t$

$= \frac{1}{2}(25 + 0) \frac{m}{s} \ 16\frac{2}{3} (s) = 208\frac{1}{3} \ m$

12. A ball is thrown vertically upward with an initial velocity of 15 m/s. Use $g = 10$ m/s^2.

(a) How long does it take to reach its maximum height?

$V = 15 \frac{m}{s} + (-10 \frac{m}{s^2}) t = 0$

$t = \frac{15 \ m/s}{10 \ m/s^2} = 1.5 \ s$

(b) What is the maximum height reached?

$d = \frac{1}{2}(15 \frac{m}{s} + 0) 1.5 s$ -

$= 11.25 \ m$

(c) What is the ball's velocity when it returns to the thrower's hand?

15 m/s downward

Symmetry